

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

Semester – I / II University Examination May 2010

B.Tech. (CE/EE/IT) / B.Tech. (CL/EC/ME)

CE – 101 Fundamentals of Computing & Programming

Date: 18/05/2010, Tuesday

Time: 12:30 p.m to 3:30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Answer the following questions.

05

1. Draw the memory allocation layout for int a[3] where base address is 2011.
2. Define C tokens.
3. Explain explicit type conversion.
4. Which are the modes of a file supported by fopen()?
5. What is the scope and life of a static variable?

Q.1(b) What is the correct output of the following code?

06

```
(1) #define M 3
void main ( )
{
    int m, N = 5;
    for (m=1 ; N!=1; )
    {
        printf ("%d\n", M);
        ++m;
        N--;
    }
}
```

```
(2) void main ( )
{
    int i;
    char c[5] = "ABCD";
    for(i=1; i<5; i++)
        c[i] = c[i]+32;
    printf ("%d\n", i);
    printf ("%3s", c);
}
```

```
(3) void main()
{
    int option=7;
    switch(option%4)
    {
        case 1: printf("ABC");
        case 2: printf("DEF");
        case 3: printf("GHI");
        case 4: printf("JKL");
    }
}
```

Q.2 Write programs in C.

12

1. Write a program to print the Armstrong number between 1 to N, where N is user dependent. (Armstrong no is if $153 = 1^3 + 5^3 + 3^3$)
2. Write a program to calculate and print the total number of vowels (a,e,i,o,u) present in small letter and capital letter separately in the entered sentence.

OR

Q.2 Write programs in C.

12

1. Write a program to convert decimal into its equivalent binary number.
2. Write a C program to calculate the electricity bill using if...elseif... as follows.
Given the number of units consumed, unit charges are as follows:

- | | |
|------------------------|---------------|
| (a) For first 50 units | Rs. 0.50/unit |
| (b) For next 100 units | Rs. 0.75/unit |
| (c) For next 100 units | Rs. 1.20/unit |
| (d) For unit above 250 | Rs. 1.50/unit |

Add fuel surcharge 20% and Govt. Tax 10% on bill to calculate the final bill.

P.T.O.

Q.3 Answer the following questions.

12

1. Differentiate: Structure vs. Union
2. Explain basic structure of a C program.
3. Explain recursion with example.

OR

Q.3 Answer the following questions.

12

1. Explain three classes of the data types. Explain primary data types in detail.
2. Differentiate between *Call by value* and *Call by address*.
3. Which are the categories of the user-defined function? Explain any one in detail.

SECTION – II

Q.4 Do as directed.

11

1. Draw the flowchart to find the maximum number from the three numbers.

02

2. Evaluate the following expression with steps according to C.

02

$$(3 + 4) \% 4 * 6 / 3 - 1$$

3. Evaluate: $(BCD)_{16} = (?)_{10}$ and $(1011\ 1011\ 1101)_2 = (?)_8$

02

4. Fill in the blanks with the most appropriate answer.

(i) '1' is an example _____ constant and "1" is an example of _____ constant.

02

(ii) _____ and _____ are optional in switch statement.

5. Classify between valid and invalid variable name in C.

03

(i) RAM

(ii) ANSI

(iii) GJ@India

(iv) Float variable

(v) One_int

(vi) ID#22

Q.5 Write programs in C.

12

1. Write a program to find whether the string is palindrome or not without *string.h*.

2. Define a structure Time having integer data members hour, minute, second. Write a program to enter and then add two variables and store the result into third variable. Validate the seconds and minutes of the result and print it.

(e.g. 3 Hour 65 Min 70 Sec = 4 Hour 6 Min 10 Sec)

OR

Q.5 Write programs in C.

12

1. Write a program to evaluate the following series using user-defined function *fact()* that returns the factorial of an argument.

$$1! - 2! + 3! - 4! + \dots \pm n!$$

2. Write a program to calculate the sum of the elements of an array of 10 float using pointer. (Use pointer to enter and to add elements)

Q.6 Answer the following questions.

12

1. Explain the functions: *getch()*, *getchar()* and *gets()*.
2. Explain the various methods to initialize the character array.
3. What is pointer? What is the scale factor of a pointer?
4. Explain short hand operators and pre/post decrement operator.

OR

Q.6 Answer the following questions.

12

1. Differentiate between local and global variable.
2. Explain actual argument and formal argument of a user-defined function.
3. Explain the function used for random access of a file contents.
4. Explain relational and sizeof operators.
